THE LOCUS WHERE A SECTION OF AN INVERTIBLE MODULE DOES NOT VANISH IS OPEN AND TRIVIALISES THE MODULE THERE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The tensor product of an invertible module with its inverse as the structure sheaf. Invoked inside the counted proof at source lines 4596-4602 as a pointer to a lemma of this chapter. Produces the pairing under which the section becomes a basis element.
- Nakayama's lemma for a local ring, applied to the stalk of a free module of rank one. Invoked inside the counted proof at source lines 4604-4606 as a cross-chapter pointer. Shows that a section not lying in the maximal ideal times the module is a basis of the stalk.

Source reference. Original lemma lemma-constructions-invertible: sources/stacks-modules/pinned-original-source.tex, source lines 4465–4465. Navigation pointer only: the counted text refers to the Stacks item lemma-constructions-invertible, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-lemma-NAK: sources/stacks-modules/pinned-original-source.tex, source lines 4605–4605. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-NAK, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Lemma 0.1. *Let X be a ringed space. Assume that each stalk $\mathcal{O}_{X,x}$ is a local ring with maximal ideal $\mathfrak{m}_x$. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. For any section $s \in \Gamma(X, \mathcal{L})$ the set*

$$X_s = \{x \in X \mid \text{image } s \notin \mathfrak{m}_x \mathcal{L}_x\}$$

is open in X. The map $s : \mathcal{O}_{X_s} \to \mathcal{L}|_{X_s}$ is an isomorphism, and there exists a section s' of $\mathcal{L}^{\otimes -1}$ over X_s such that $s'(s|_{X_s}) = 1$.

Proof. Suppose $x \in X_s$. We have an isomorphism

$$\mathcal{L}_x \otimes_{\mathcal{O}_{X,x}} (\mathcal{L}^{\otimes -1})_x \longrightarrow \mathcal{O}_{X,x}$$

by Lemma lemma-constructions-invertible. Both $\mathcal{L}_x$ and $(\mathcal{L}^{\otimes -1})_x$ are free $\mathcal{O}_{X,x}$-modules of rank 1. We conclude from Algebra, Nakayama's Lemma algebra-lemma-NAK that s_x is a basis for $\mathcal{L}_x$. Hence there exists a basis element $t_x \in (\mathcal{L}^{\otimes -1})_x$ such that $s_x \otimes t_x$ maps to 1. Choose an open neighbourhood U of x such that t_x comes from a section t of $\mathcal{L}^{\otimes -1}$ over U and such that $s \otimes t$ maps to $1 \in \mathcal{O}_X(U)$. Clearly, for every $x' \in U$ we see that s generates the module $\mathcal{L}_{x'}$. Hence $U \subset X_s$. This proves that X_s is open. Moreover, the section t constructed over U above is unique, and hence these glue to give the section s' of the lemma. □